

Kryštof Przeczek

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PROFILE

I am a Computer Science student focused on graphics programming, real-time rendering, and simulation. I have experience building Vulkan renderers, path tracers, and physics simulations from scratch in C/C++. I have an interest in performance-critical systems combining computer science, mathematics, physics, and visual art.

SKILLS

- **Languages:** C, C++, C#, GLSL, HLSL, Slang (rasterisation and compute tasks), Java, and Python.
- **Graphics APIs:** OpenGL and Vulkan.
- **Tools:** Git, Unix terminal, RenderDoc, Xcode, and Radeon Developer Suite.
- **Engines & Frameworks:** C# FNA/MonoGame library, Unity, Godot, and GameMaker Studio 2.

PROJECTS

Magpie | C, SDL, Vulkan, ImGui | <https://github.com/kryzp/magpie>

Vulkan renderer with hot-reloadable code and assets, async asset streaming, a fiber-based job system, physically-based materials, Lua scripting integration, compute culling, skeletal animation support, and a render graph and scene system for GPU synchronization and geometry management.

Stick it Together | FastAPI, Kotlin, React, ElevenLabs | <https://github.com/kryzp/sotonhack26backend>

Winner of SotonHack '26. 1v1 party game linking concepts via semantic similarity scoring.

Path Tracer | C++ | <https://github.com/kryzp/pathtracer>

Path tracer capable of physically-based metallic, lambertian, and refractive materials.

2D Finite-Element Solver | Java | <https://github.com/kryzp/simple-finite-element-solver>

Computes beam deflection via stiffness matrices built from a 2D structure and boundary conditions.

2D Navier-Stokes Fluid Solver | C++ | <https://github.com/kryzp/simple-navier-fluid-sim>

Fluid simulation using iterative projection methods to solve for diffusion, advection and external forces.

WORK EXPERIENCE

Microsoft

June 2022–July 2022

Summer Work Placement

Shadowed a senior telecommunications software engineer and learned about the tools and algorithms used daily to ensure the rapid response times in their high-speed communications network.

Southampton Data Science Society

2025–2026

Social Media Manager

Worked on managing the social media of and promoting the data science society. Made post graphics and logo.

EDUCATION

University of Southampton

2024–Present

Computer Science MEng

Hills Road Sixth Form College

2022–2024

Further Maths A, Maths A, Computer Science A, Physics B

Cambourne Village College

2018–2022

GCSE's with 9's in Further Maths, Maths and Physics

ACHIEVEMENTS

- Competed in Advent of Code, placing 1st in the school (out of 150) in Year 13, and 2nd in Year 12.
- Earned Silver in both the British Informatics Olympiad (BIO) and British Physics Olympiad (BPhO).
- Completed 104 Project Euler problems independently, ranking me in the top 5% of all competitors.
- Duke of Edinburgh's Award: Refined skills in game development, swimming, and volunteering at a local library.

HOBBIES

- Analysing games by intercepting CPU-to-GPU data to study their graphics techniques.
- Experimenting with esoteric shaders on Shadertoy to create visually striking scenes.
- Sketching landscapes and painting wildlife digitally.
- Playing guitar, I'm a fan of post-rock and post-apocalyptic styles.
- Undertaking more hands-on projects like building a crossbow and carpentry.